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Prop descomposicion ortogonal

Proposición 1. *Sea H un espacio de Hilbert y $M \subset H$ un subespacio cerrado y $x \in H$, entonces $x = P_M(x) + P_{M^\perp}(x)$ y esta descomposición es única.*

***Demostración*:** Sea $v \in M^\perp$, entonces $\langle x - (x - P_M(x)), v \rangle = \langle P_M(x), v \rangle = 0$ ya que $P_M(x) \in M$. Por tanto, por el `teo-carac-proyeccion-ortogonal-subespacio-cerrado/`**??**, $x - P_M(x) = P_{M^\perp}(x)$. ■

Referenciado en

- `teo-proyeccion-ortogonal`

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